ASSIGNMENT 1

NAME : VANSH JINDAL

UNIV ROLL NO : 2415001735

SECTION: SB

CLASS ROLL NO: 63

SESSION: 2024-2028

SUBMITTED TO : Mr AKASH SIR

Instructions for Students

- Create **separate `.js` files** for each question.
- Use **comments** to explain your logic.
- Test all programs using the **browser console** or **Node.js**.
- Use **descriptive variable names**, indentation, and formatting.
- Use only **core JavaScript** (no frameworks or libraries).

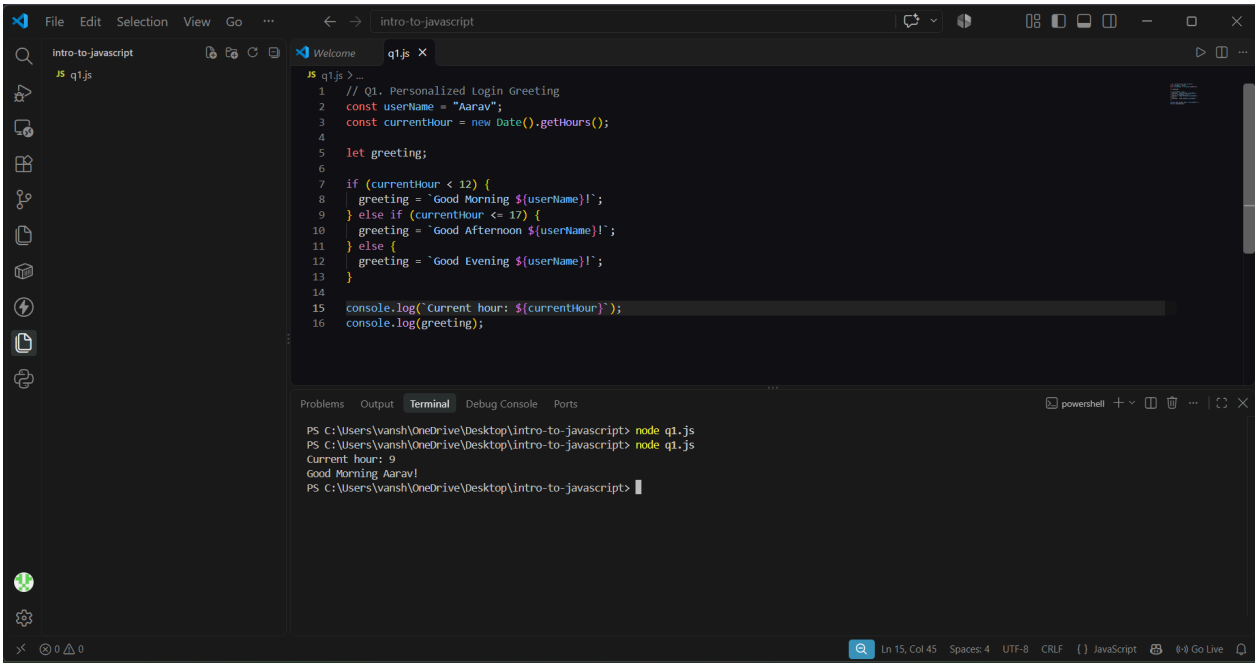
Q1. Personalized Login Greeting

Create a program that greets users with time-based messages.

Requirements:

- Declare variables for `userName` and current hour using `new Date().getHours()`.

- If time is before 12 → “Good Morning [name]!”
- If time is between 12 and 17 → “Good Afternoon [name]!”
- Otherwise → “Good Evening [name]!”
- Use template literals and proper string formatting.

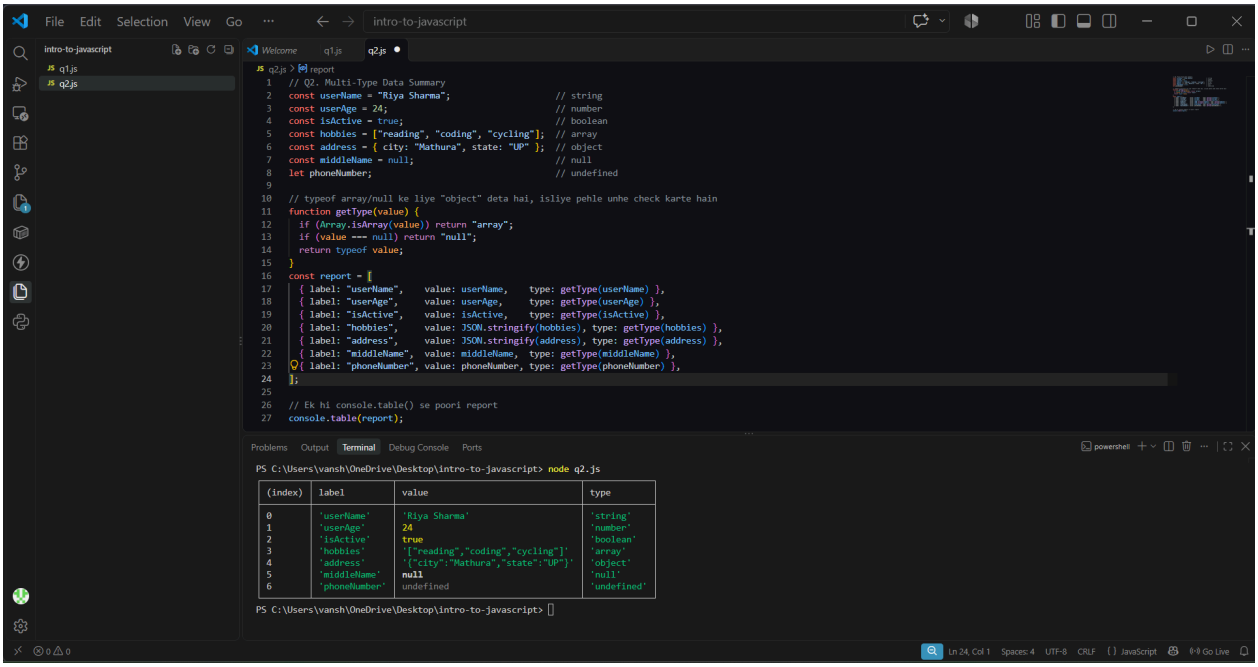


Q2. Multi-Type Data Summary

Simulate a database summary of various user data entries.

Requirements:

- Declare variables of different data types: string, number, boolean, array, object, null, undefined.
- Use `typeof` operator and `Array.isArray()` to identify each type. ● Print a formatted report (label + value + type) using one `console.table()` statement.

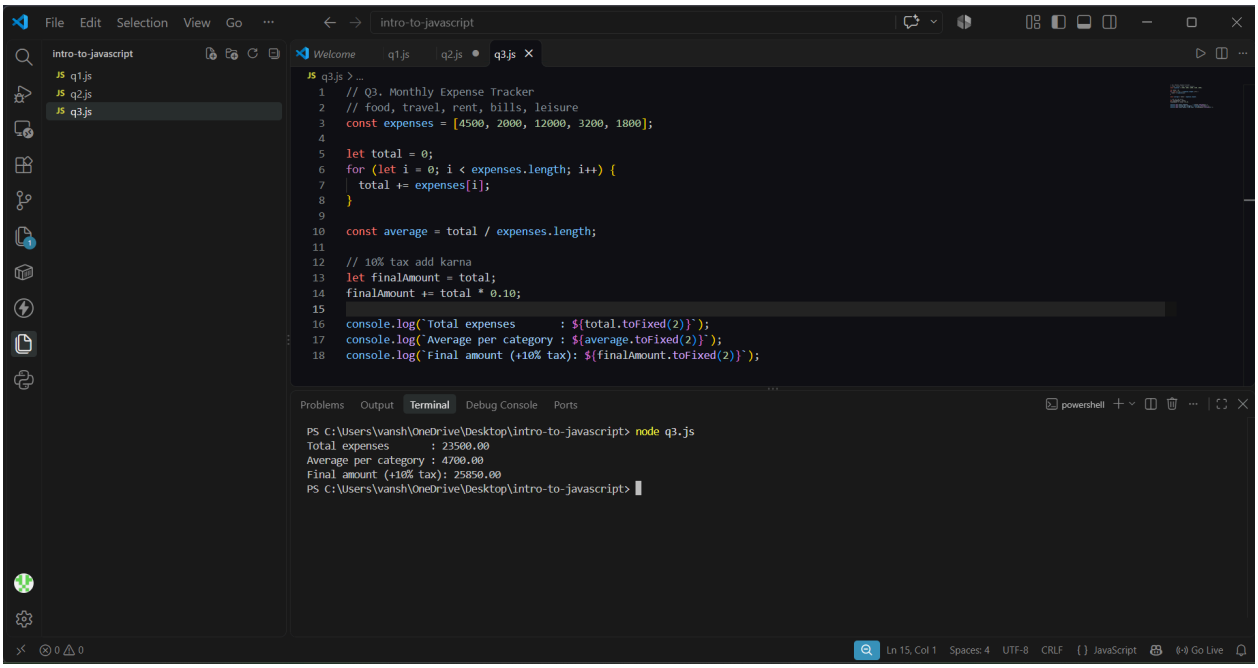


Q3. Monthly Expense Tracker

Develop a script that calculates total and average monthly expenses with dynamic updates.

Requirements:

- Use an array of expenses for 5 categories (food, travel, rent, bills, leisure).
- Use arithmetic and assignment operators to add 10% tax to the total. ● Use `toFixed(2)` to round values.
- Display total, average, and final amount after tax.

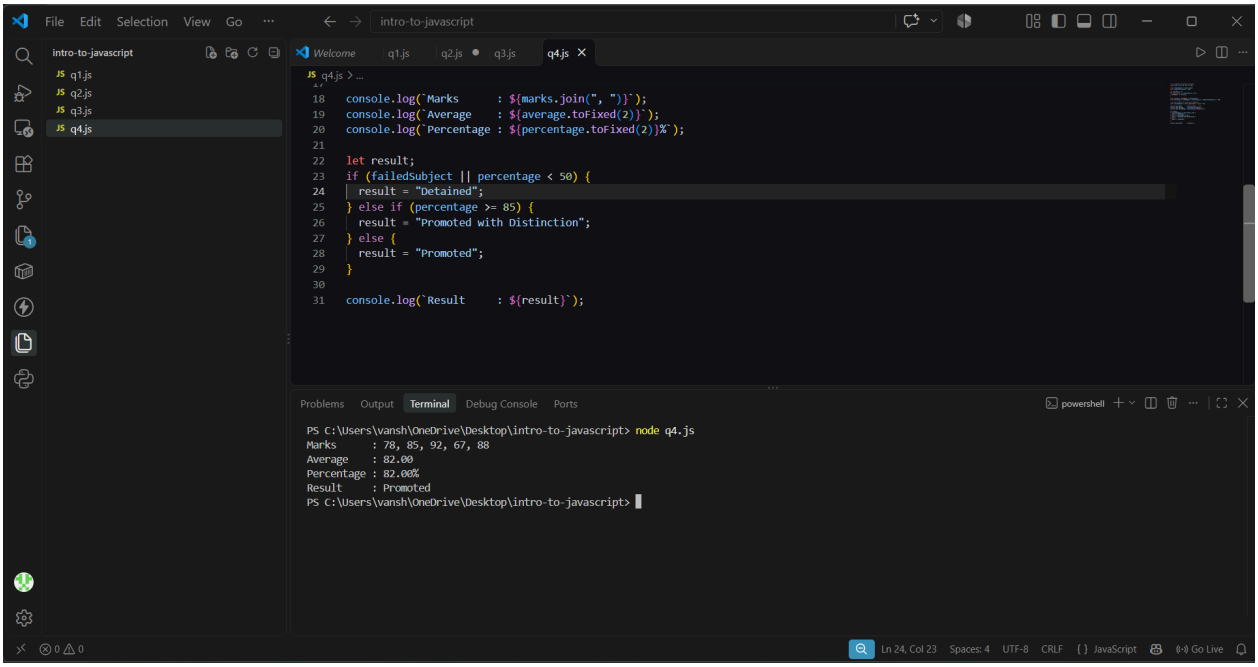


Q4. Academic Performance Evaluator

Check whether a student is promoted or not based on multiple subjects.

Requirements:

- Input marks of 5 subjects (use an array).
- Calculate average and overall percentage.
- Use logical operators to print:
    - “Promoted with Distinction” if ≥ 85%
    - “Promoted” if between 50%–84%
    - “Detained” if < 50%
- Add validation: if any subject < 35 → automatically “Detained”.

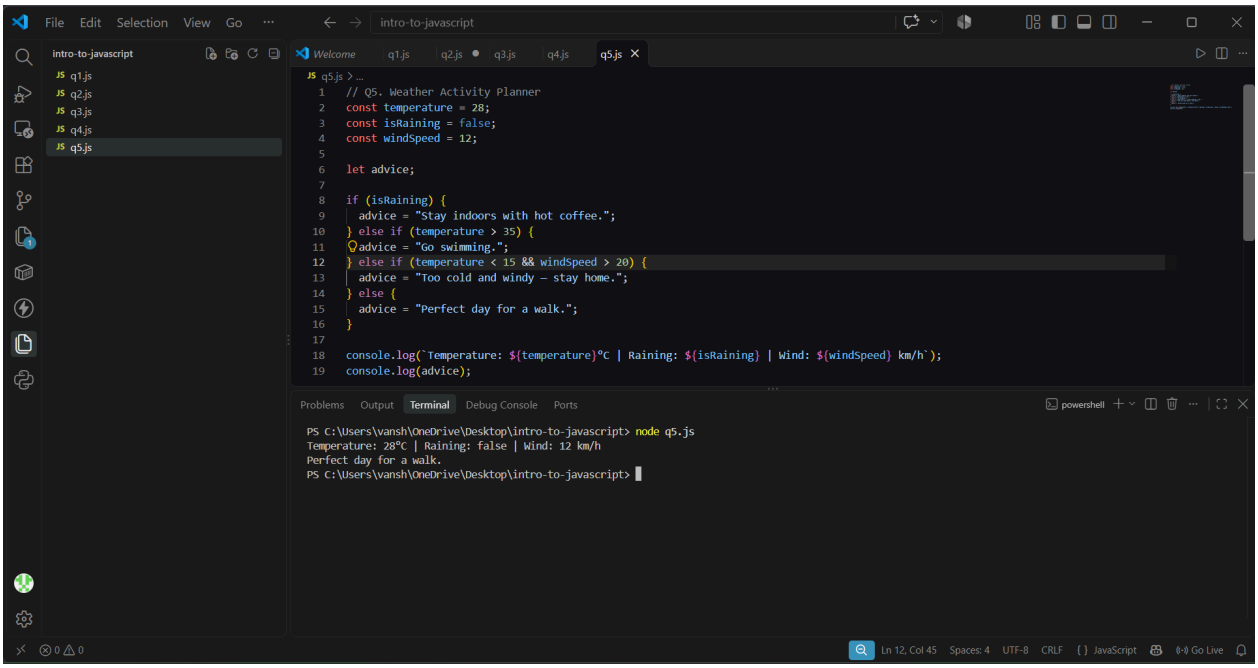


Q5. Weather Activity Planner

Advise a user on what to do based on temperature, rain, and wind speed.

Requirements:

- Create variables for `temperature`, `isRaining`, `windSpeed`.
- If raining → “Stay indoors with hot coffee.”
- Else if temperature > 35 → “Go swimming.”
- Else if temperature < 15 and windSpeed > 20 → “Too cold and windy — stay home.”
- Else → “Perfect day for a walk.”
- Combine && and || conditions smartly.

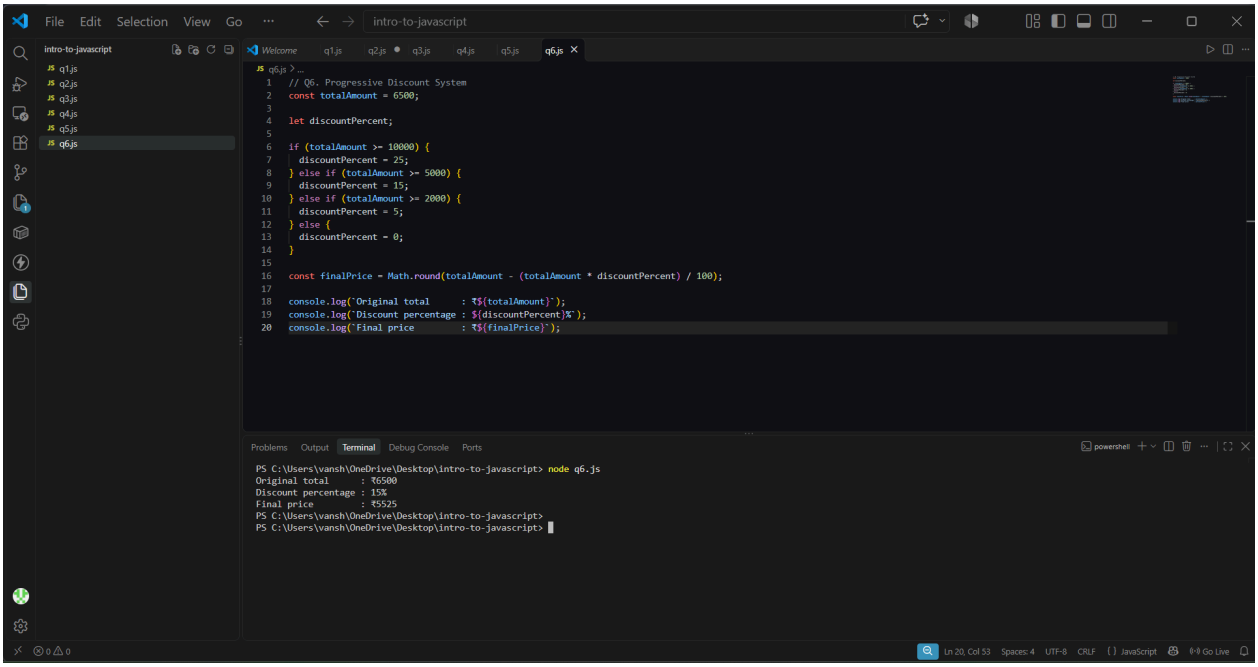


Q6. Progressive Discount System

Build a system that applies dynamic discount tiers for shopping totals. **Requirements:**

- Input total purchase amount.
- If total ≥ 10000 → 25% discount.
- Else if ≥ 5000 → 15% discount.
- Else if ≥ 2000 → 5% discount.
- Otherwise → no discount.
- Calculate and print:
    - Original total
    - Discount percentage
    - Final price after discount

- Use `Math.round()` for rounding.



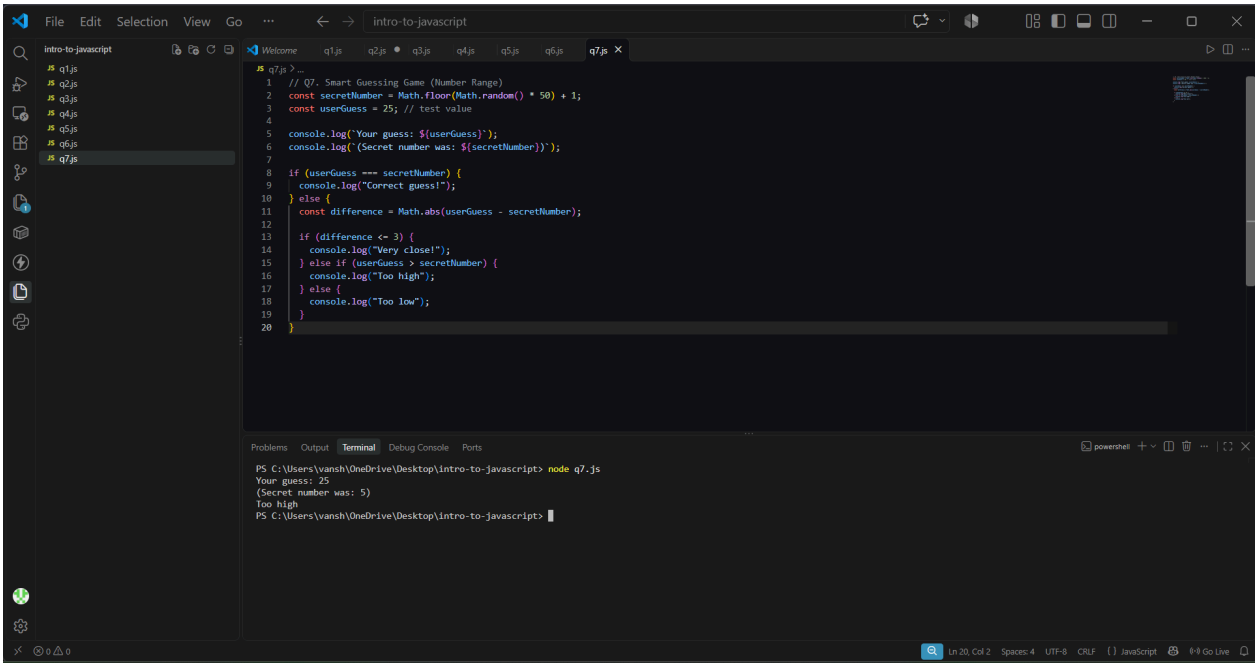
Q7. Smart Guessing Game (Number Range)

Enhance the guessing game by allowing range checking.

Requirements:

- Generate a random secret number between 1–50 using `Math.floor(Math.random() * 50) + 1`.
- Let `userGuess` be a variable with a test value.
- Print:
    - “Correct guess!” if equal
    - “Very close!” if within ± 3 of secret
    - “Too high” or “Too low” otherwise

- Use nested `if` conditions and logical operators.



Q8. Employee Salary Projection

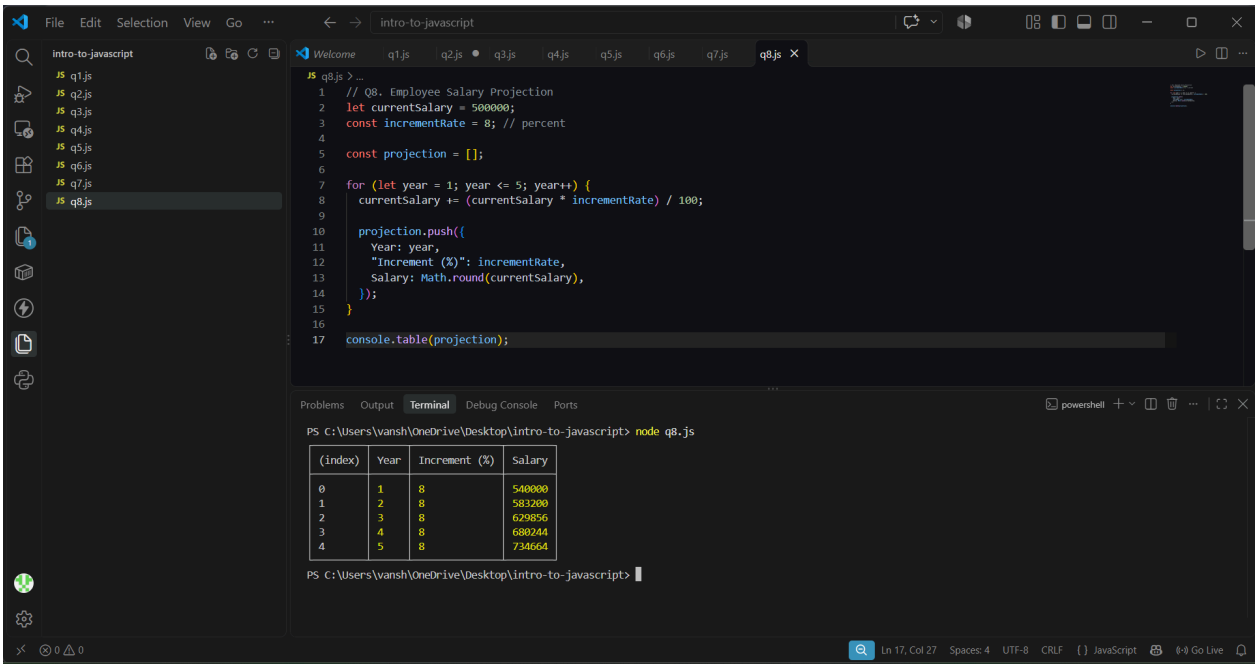
Calculate a 5-year salary projection with yearly increments.

Requirements:

- Declare current salary and annual increment rate (in %).
- Use a `for` loop to compute salary for each of 5 years.

- Print yearly salary in a formatted console table.

- Apply rounding and assignment operators.

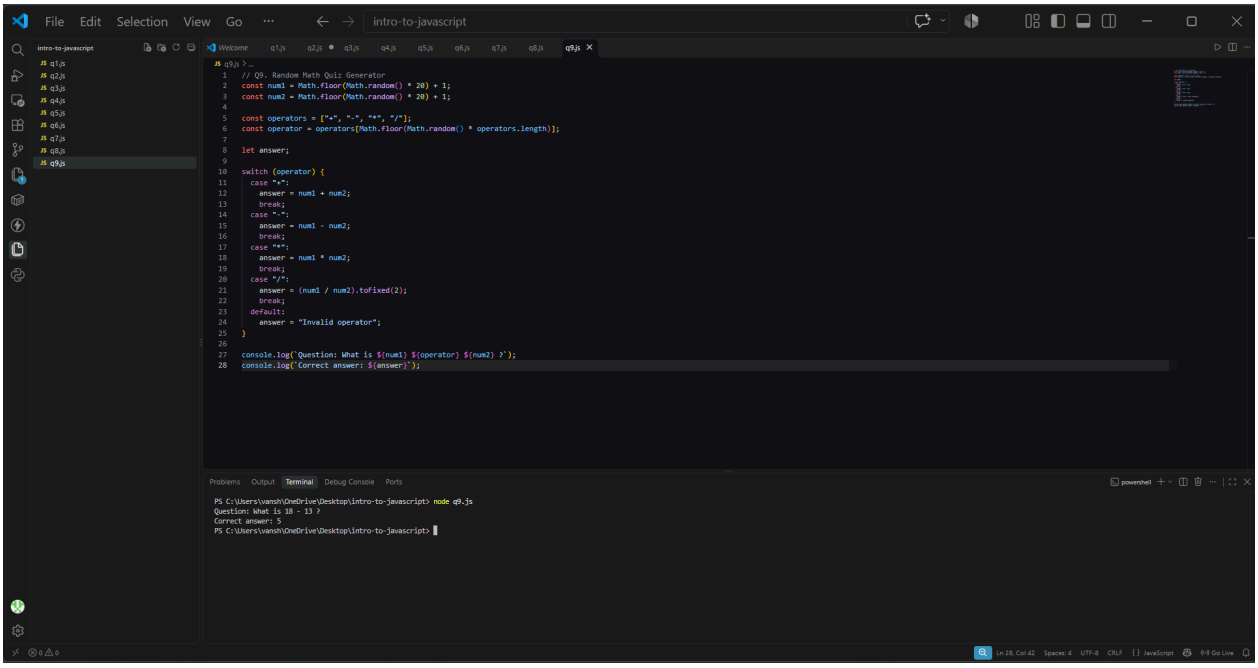


Q9. Random Math Quiz Generator

Generate random arithmetic challenges and evaluate user answers automatically.

Requirements:

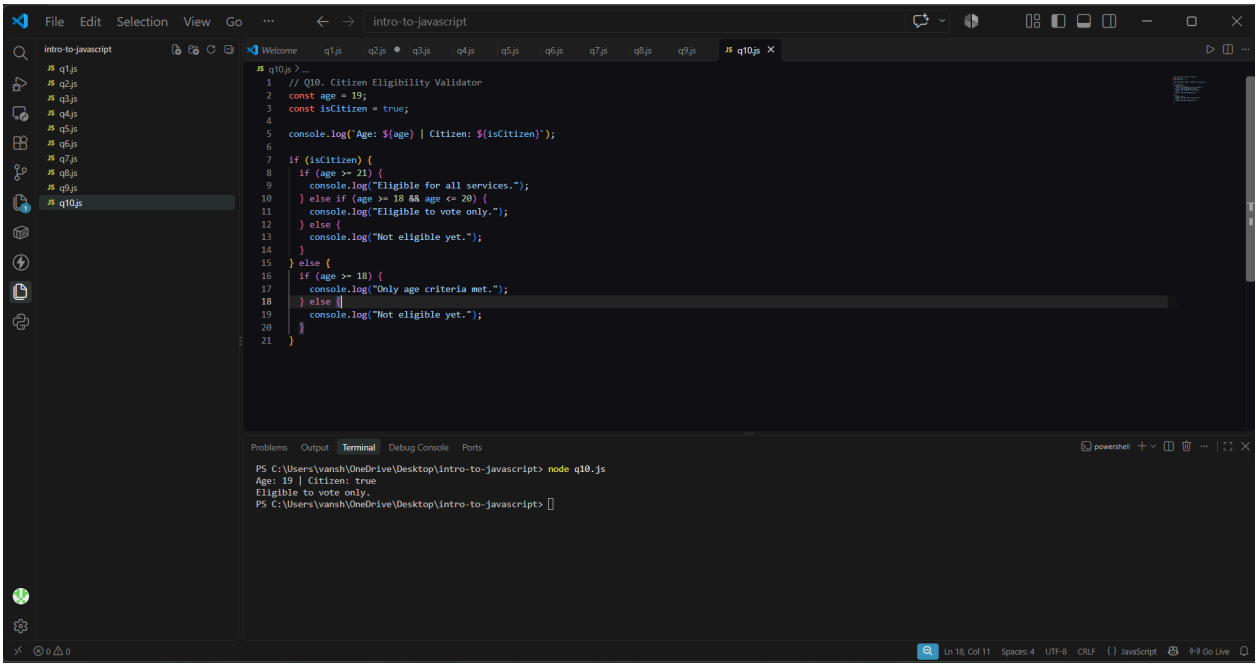
- Use `Math.random()` to generate two numbers (1–20) and a random operator from `['+', '-', '*', '/']`.
- Calculate the correct answer using `switch` statement.
- Display question and correct answer in the console.
- Ensure division is rounded to two decimals with `toFixed(2)`.



Q10. Citizen Eligibility Validator

Evaluate if a person can vote, drive, and apply for a passport based on age and citizenship. **Requirements:**

- Input: `age` and `isCitizen` (boolean).
- Use nested `if-else` and logical operators to print:
    - “Eligible for all services.” if citizen and age ≥ 18
    - “Eligible to vote only.” if citizen and 18–20
    - “Only age criteria met.” if not citizen but age ≥ 18
    - “Not eligible yet.” otherwise
- Use `console.log()` for all outcomes.


File Edit Selection View Go
intro-to-javascript
intro-to-javascript
q1.js
q2.js
q3.js
q4.js
q5.js
q6.js
q7.js
q8.js
q9.js
q10.js
Welcome
q1.js
q2.js
q3.js
q4.js
q5.js
q6.js
q7.js
q8.js
q9.js
q10.js
q10.js > ...
1 // Q10. Citizen Eligibility Validator
2 const age = 19;
3 const isCitizen = true;
4
5 console.log(`Age: ${age} | Citizen: ${isCitizen}`);
6
7 if (isCitizen) {
8 if (age >= 21) {
9 console.log("Eligible for all services.");
10 } else if (age >= 18 && age <= 20) {
11 console.log("Eligible to vote only.");
12 } else {
13 console.log("Not eligible yet.");
14 }
15 } else {
16 if (age >= 18) {
17 console.log("Only age criteria met.");
18 } else {
19 console.log("Not eligible yet.");
20 }
21 }
Problems
Output
Terminal
Debug Console
Ports
powershell
PS C:\Users\vansh\OneDrive\Desktop\intro-to-javascript> node q10.js
Age: 19 | Citizen: true
Eligible to vote only.
PS C:\Users\vansh\OneDrive\Desktop\intro-to-javascript>
Ln 18, Col 11
Spaces: 4
UTF-8
CRLF
JavaScript
Go Live